MATH 540. HW #12.

Prob 1 (2&T. Ch. 7.)

(a) Carefully write a "well-posed" Neumann pr-m for the Laplace eq n. Why is "well-posed" in quotes?

Sol-n.

In order for a PDE to be well-posed:

(i) sol-n exists

(ii) The sol-n is unique

(iii) The sol-n continuously depends on data.

So, consider

(1) $\Delta u = 0$ on Ω

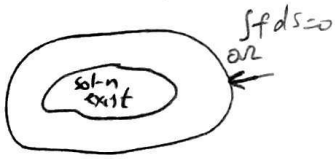
(2) $\frac{\partial u}{\partial \underline{n}} = f$ on $\partial\Omega$, $f \in C^0(\partial\Omega)$,

Let me state for (1)+(2) to be "well-posed" conditions:

(i) By the ↓ Thm 1

(3) $\boxed{\int_{\partial\Omega} f\,ds = 0}$ is necessary for a sol-n to exist to the P.P, (1)+(2).

i.e. If $\int_{\partial\Omega} f\,ds \neq 0$, then sol-n does not exist.

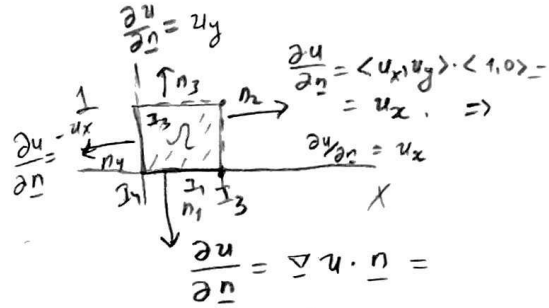


(ii) By the ↓ Thm 2,

Any 2 sol-ns of the Neumann pr-m, (1)+(2), <u>differ by a constant.</u>

(iii) By (2), we can assess, the sol-n continuously depends on f, so by relaxing the <u>uniqueness condition</u> (because the sol-n of Neumann pr-m is not unique, differs by a constant from one another) & satisfying (i) & (ii), we could say Neumann pr-m (1)+(2) is "well-posed". The reason for the quote is to mean uniqueness is guaranteed up to a constant. ∎

(b) Let Ω be the unit square, $0 < x < 1$, $0 < y < 1$. Give 2 examples of Neumann bdry conditions for which we are <u>not</u> guaranteed at least one sol-n. Explain why we are not guaranteed at least one sol-n.



$$\frac{\partial u}{\partial n} = \nabla u \cdot \underline{n} = \langle u_x, u_y \rangle \cdot \langle 0, -1 \rangle = -u_y \Rightarrow$$

$$\frac{\partial u}{\partial n} = -u_y$$

Sol-n.

Let

(1) $$\Omega = \{(x,y) : 0 < x < 1,\ 0 < y < 1\}$$

(*) On 4 sides of the unit rectangle, we have 4 different outward normal vectors n_1, n_2, n_3, n_4. So, for each side the $\frac{\partial u}{\partial \underline{n}}$ is prescribed separately.

(**) From part (a), remember the necessary cond-n for a sol-n to exist:

(2) $$\int_{\partial\Omega} f \, ds = 0.$$

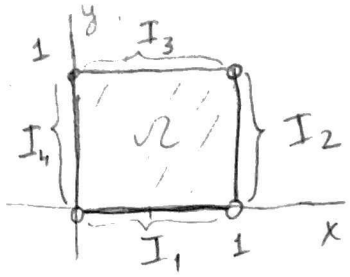
Hence, to be not guaranteed at least one sol-n (i.e. there does NOT exist a sol-n), we must have

(3) $$\int_{\partial\Omega} f \, ds \neq 0.$$

- So considering (*) & (3), we need to prescribe $\frac{\partial u}{\partial \underline{n}}$, s.t. the total $\int_{\partial\Omega} f ds = \int_{I_1} f ds + \int_{I_2} f ds + \int_{I_3} f ds + \int_{I_4} f ds \neq 0$.

Then consider the following way of giving $\frac{\partial u}{\partial \underline{n}}$:

(i) First example:



$$(4)\quad \frac{\partial u}{\partial \underline{n}} = f(x,y) = \begin{cases} x, & I_1 = \{(x,y):\ 0 \le x \le 1,\ y = 0\} \\ y+1, & I_2 = \{(x,y):\ x = 1,\ 0 \le y \le 1\} \\ -2x+2, & I_3 = \{(x,y):\ 0 \le x \le 1,\ y = 1\} \\ 0, & I_4 = \{(x,y):\ x = 0,\ 0 \le y \le 1\} \end{cases}$$

Note that first $f(\underline{x}) = f(x,y)$ defined in (4) satisfies

$$(5)\qquad f \in C^0(\partial\Omega),$$

but we can <u>argue</u> (claim), the Laplace eq-n $\Delta u = 0$ with the prescribed condition in (4) does <u>NOT</u> have a sol-n since

$$(6)\qquad \int_{\partial\Omega} f(x,y)\,ds = \int_{I_1} f\,ds + \int_{I_2} f\,ds + \int_{I_3} f\,ds + \int_{I_4} f\,ds \neq 0 .$$

To verify (6), let's see the ↓ calcu-n:

$$\int_{\partial\Omega} f(x,y)\,ds = \int_{I_1} x\,dx + \int_{I_2} (y+1)\,dy + \int_{I_3} (-2x+2)\,dx + \int_{I_4} 0\,dy =$$

$$= \int_0^1 x\,dx + \int_0^1 (y+1)\,dy + \int_0^1 (-2x+2)\,dx + \int_0^1 0\,dy =$$

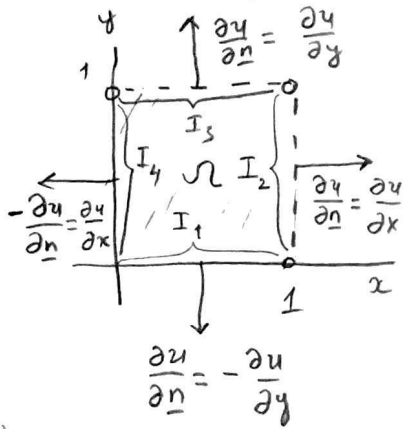
$$= \left[\frac{x^2}{2}\right]_{x=0}^{1} + \left[\frac{y^2}{2} + y\right]_{y=0}^{1} + \left[-x^2 + 2x\right]_{x=0}^{1} + 0 = \left[\frac{1}{2}\right] + \left[\frac{1}{2} + 1\right] + \left[-1+2\right] + 0 =$$

$$= 1 + 2 = 3.$$ That is

$$(7)\qquad \int_{\partial\Omega} f(\underline{x})\,ds = 3 \neq 0 \Rightarrow$$ There's NO sol-n for $\Delta u = 0$ with (4).

(ii) Second example:

We can try modifying the prescription in example (i), in (4), keeping the idea the same:



(8) $$\frac{\partial u}{\partial \underline{n}} = \begin{bmatrix} -u_y \\ u_x \\ u_y \\ -u_x \end{bmatrix} = f(x,y) = \begin{cases} -x^2, & I_1 = \{(x,y):\ 0 \le x \le 1,\ y = 0\} \\ y-1, & I_2 = \{(x,y):\ x = 1,\ 0 \le y \le 1\} \\ 2x, & I_3 = \{(x,y):\ 0 \le x \le 1,\ y = 1\} \\ 2-2y, & I_4 = \{(x,y):\ x = 0,\ 0 \le y \le 1\} \end{cases}$$

Again note that $f(\underline{x}) = f(x,y)$ defined in (4) satisfies

(9) $$f \in C^0(\partial\Omega),$$

But we claim, the Laplace eq-n $\Delta u = 0$ with the prescribed <u>Neumann boundary</u> condition in (8) does NOT have a sol-n since

(10) $$\int_{\partial\Omega} f(x,y)\,ds = \int_{I_1} f\,ds + \int_{I_2} f\,ds + \int_{I_2} f\,ds + \int_{I_4} f\,ds \neq 0.$$

To verify (10), consider the ↓ calcul-n:

$$\int_{\partial\Omega} f(x,y)\,ds = \int_{I_1} -x^2\,dx + \int_{I_2} (y-1)\,dy + \int_{I_3} 2x\,dx + \int_{I_4} (2-2y)\,dy =$$

$$= \int_0^1 -x^2\,dx + \int_0^1 (y-1)\,dy + \int_0^1 2x\,dx + \int_0^1 (-2y+2)\,dy =$$

$$= \left[-\frac{x^3}{3}\right]_{x=0}^{1} + \left[\frac{y^2}{2} - y\right]_{y=0}^{1} + \left[x^2\right]_{x=0}^{1} + \left[-y^2 + 2y\right]_{y=0}^{1} =$$

$$= \left[-\frac{1}{3}\right] + \left[\frac{1}{2} - 1\right] + [1] + [-1+2] = -\frac{1}{3} + \frac{1}{2} + 1 = \frac{7}{6} \neq 0.$$ That is

$\int_{\partial\Omega} f(\underline{x})\,ds = \frac{7}{6} \neq 0 \Rightarrow$ There's NO sol-n for $\Delta u = 0$ with the Neumann BC in (8).

(c) Let Ω be the unit square,

(1) $$\Omega = \{(x,y):\ 0 < x < 1,\ 0 < y < 1\}.$$

Give 2 examples of Neumann boundary cond-ns for which we are guaranteed at least one sol-n. Explain.

Sol-n:

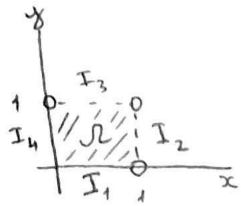
[Preamble].

To be guaranteed to <u>have at least one</u> sol-n, we can try to modify the Neumann BC prescribed in part (b), (4) & (8) so that

(2) $$\int_{\partial\Omega} f(\underline{x})\, ds = 0 \quad \& \quad f \in C^0(\partial\Omega).$$

(i) First example, consider the ↓ BC as an easy example:

(3) $$\frac{\partial u}{\partial \underline{n}} = \begin{bmatrix} -\frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \\ -\frac{\partial u}{\partial x} \end{bmatrix} = f(x,y) = \begin{cases} 0 \in \mathbb{R}, & I_1 = \{(x,y):\ 0 \le x \le 1,\ y = 0\} \\ 0, & I_2 = \{(x,y):\ x = 1,\ 0 \le y \le 1\} \\ 0, & I_2 = \{(x,y):\ 0 \le x \le 1,\ y = 1\} \\ 0, & I_4 = \{(x,y):\ x = 0,\ 0 \le y \le 1\} \end{cases}$$



- First, note that $f(\underline{x}) = f(x,y) = 0$, defined in (3), satisfies

(4) $$f(\underline{x}) = a \in C^0(\partial\Omega)$$

- Second,

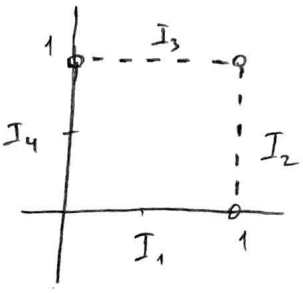
(5) $$\int_{\partial\Omega} f(x,y)\, ds = \int_{I_1} f\, ds + \int_{I_2} f\, ds + \int_{I_3} f\, ds + \int_{I_4} f\, ds = 0,$$ because

$$\int_{\partial\Omega} f(x,y)\, ds = \int_{I_1} 0\, ds + \int_{I_2} 0\, ds + \int_{I_3} 0\, ds + \int_{I_4} 0\, ds = 0 + 0 + 0 + 0 = 0.$$ (All integrands are = 0) $\Rightarrow$

(6) $$\int_{\partial\Omega} f(\underline{x})\, ds \equiv 0.$$

Then by necessary Thm for existance of a sol-n, there <u>is</u> a sol-n, confirming the pysical intiution for $f = 0$, being perfectly insulated case.

(ii) Example two. Consider the ↓ Neuman BC :



$$(7)\quad \frac{\partial u}{\partial \underline{n}} = \begin{bmatrix} -u_y \\ u_x \\ u_y \\ -u_x \end{bmatrix} = f(x,y) = \begin{cases} x, & I_1 = \{(x,y):\ 0 \le x \le 1,\ y=0\} \\ 1-y, & I_2 = \{(x,y):\ x=1,\ 0 \le y \le 1\} \\ -x, & I_3 = \{(x,y):\ 0 \le x \le 1,\ y=1\} \\ -1+y, & I_4 = \{(x,y):\ x=0,\ 0 \le y \le 1\} \end{cases}$$

- First note that

$$(8)\qquad f(\underline{x}) = f(x,y) \in C^0(\partial\Omega).$$

- Second,

$$(9)\quad \int_{\partial\Omega} f(\underline{x})\,ds = \int_{I_1} f(\underline{x})\,ds + \int_{I_2} f(\underline{x})\,ds + \int_{I_3} f(\underline{x})\,ds + \int_{I_4} f(\underline{x})\,ds = 0 \quad , \quad \text{since}$$

$$\int_{\partial\Omega} f(\underline{x})\,ds = \int_{I_1} x\,dx + \int_{I_2} (1-y)\,dy + \int_{I_3} -x\,dx + \int_{I_4} (-1+y)\,dy =$$

$$= \int_0^1 x\,dx + \int_0^1 (1-y)\,dy - \int_0^1 x\,dx - \int_0^1 (y-1)\,dy =$$

$$= \underbrace{\left[\frac{x^2}{2}\right]_{x=0}^{1}}_{A} + \underbrace{\left[y - \frac{y^2}{2}\right]_{y=0}^{1}}_{B} - \underbrace{\left[\frac{x^2}{2}\right]_{x=0}^{1}}_{A} - \underbrace{\left[\frac{y^2}{2} - y\right]_{y=0}^{1}}_{B} \equiv 0 \checkmark$$

. That's

$$\int_{\partial\Omega} f(\underline{x})\,ds = 0.$$

Then, by sol-n-existance-necessary thm, there <u>does</u> exist a sol-n, confirming the physical intuition that if Ⓐ&Ⓑ amount of heat (energy) is <u>coming in</u> and Ⓐ&Ⓑ amount of heat is <u>going out</u>, then, of course, there must exist a steady-state sol-n inside the system (i.e. $\Delta u - \overset{=0}{\widetilde{u_t}} = 0 \Rightarrow \Delta u = 0$ must satisfy), assuming a good condition for the system. ∎.